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SMART SOLAR PANEL CLEANER

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING)

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Chapter 1: Introduction

Solar energy is one of the most promising renewable energy sources available today. Solar panels are widely deployed across residential, commercial, and industrial sectors to harness clean and sustainable power. However, the efficiency of solar panels significantly decreases due to the accumulation of dust, dirt, bird droppings, and other pollutants on the surface. Studies show that unclean solar panels can experience efficiency losses of up to 20-30% depending on environmental conditions.

Traditional cleaning methods are often manual, labor-intensive, water-consuming, and not cost-effective. This crates the need for an automated, intelligent, and efficient cleaning system that can maintain solar panel efficiency with minimal human intervention.

The proposed project, Smart Solar Panel Cleaner, aims to design and develop an automated cleaning mechanism integrated with sensors, robotics, and IoT-based monitoring. The system will ensure maximum solar energy utilization, reduce operational costs, and contribute to sustainable energy generation.

Chapter 2: Brief Literature Survey

Researchers and industries have explored different cleaning approaches:

- Manual cleaning: Effective but labor-intensive and costly for large installations.
- Water-based cleaning systems: Widely used but waste excessive water, unsuitable for arid regions.
- Electrostatic cleaning methods: Use electrostatic forces to repel dust but are still in experimental stages.
- Robotic cleaning systems: Mobile robots with brushes or air blowers, though expensive and requiring complex maintenance.
- IoT-based monitoring: Recent research shows integration of IoT for predictive cleaning schedules improves efficiency.

From the survey, it is evident that a cost-effective, smart, automated system is still lacking for small and medium-scale solar power plants. This project bridges that gap by proposing a hybrid design combining robotics and smart monitoring.

Several studies and existing technologies have explored ways to address the issue of solar panel
cleaning:

A) Manual Cleaning Methods: Widely used but inefficient and labor-intensive, especially in remote or large installations.

B) Robotic Cleaning Systems: Some robotic solutions have been developed but are often expensive and not scalable.

C) Water-Based Cleaning: Effective but not sustainable in regions with water scarcity.

D) Sensor-Based Systems: Incorporating dust or light sensors to monitor performance degradation due to dust.

Recent research highlights the use of automation and IoT technologies to create

intelligentsystems capable of cleaning panels without human intervention. Still, many proposed systems are either cost-prohibitive or not optimized for energy and water efficiency

Chapter 3: Problem Formulation

The major problems identified are:

- Loss of efficiency (up to 30%) due to unclean panels.
- High maintenance and water usage in traditional methods.
- Lack of affordable automation for small/medium solar users.

Hence, the need and significance of the proposed work lies in:

- Developing a low-cost, smart, eco-friendly automated system.
- Ensuring real-time monitoring of dust accumulation using sensors/IoT.
- Minimizing human labor and water wastage.

The performance of solar panels deteriorates due to:

A) Accumulation of dust and pollutants.

B) Lack of regular maintenance and cleaning.

C) High cost and labor requirements for manual or semi-automated systems. Thus, there is a need to formulate a solution that is:

D) Automated and sensor-controlled.

E) Cost-effective and easy to deploy. Water-efficient or capable of dry cleaning.

F) Smart, with remote monitoring and control capabilities.

Chapter 4: Objectives

The objectives of this project are:

1. To design and implement a smart robotic cleaning mechanism for solar panels.
2. To develop a sensor-based monitoring system for detecting dust accumulation.
3. To integrate IoT features for remote monitoring and control.
4. To evaluate the efficiency improvement after cleaning.
5. To provide a low-cost and sustainable solution for small and medium solar power plants.

Chapter 5: Methodology / Planning of Work

The proposed work will be carried out in the following phases:

1. Research & Literature Review — Study existing technologies, identify gaps.
2. Design Phase — Mechanical design of cleaning mechanism (rotating brushes, air/water spray, wiper).
3. Electronics & Sensor Integration — Using Arduino/Raspberry Pi, IR sensors, dust sensors, and IoT modules.
4. Prototype Development — Fabrication of a working model with robotic cleaning arms.
5. Testing & Optimization — Evaluate on solar panels under different environmental conditions.
6. Analysis & Documentation — Compare panel efficiency before and after cleaning.

Block Diagram (Planned):

- Solar Panel —• Dust Sensor —› Microcontroller —• Motor/Brush Mechanism —› IoT Monitoring (Mobile App/Dashboard).

Chapter 6: Facilities Required

Hardware:

- Solar panel test setup
- DC motors/servo motors
- Arduino/Raspberry Pi controller
- Dust/IR sensors
- IoT modules (ESP8266/NodeMCU)
- Battery/Power supply
- Mechanical frame with cleaning brushes/wipers

Software:

- Arduino IDE
- IoT dashboard (Blynk/Thingspeak)
- MATLAB/Python for data analysis

Other facilities:

- Lab tools for fabrication
- Testing environment with solar simulator or outdoor exposure

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